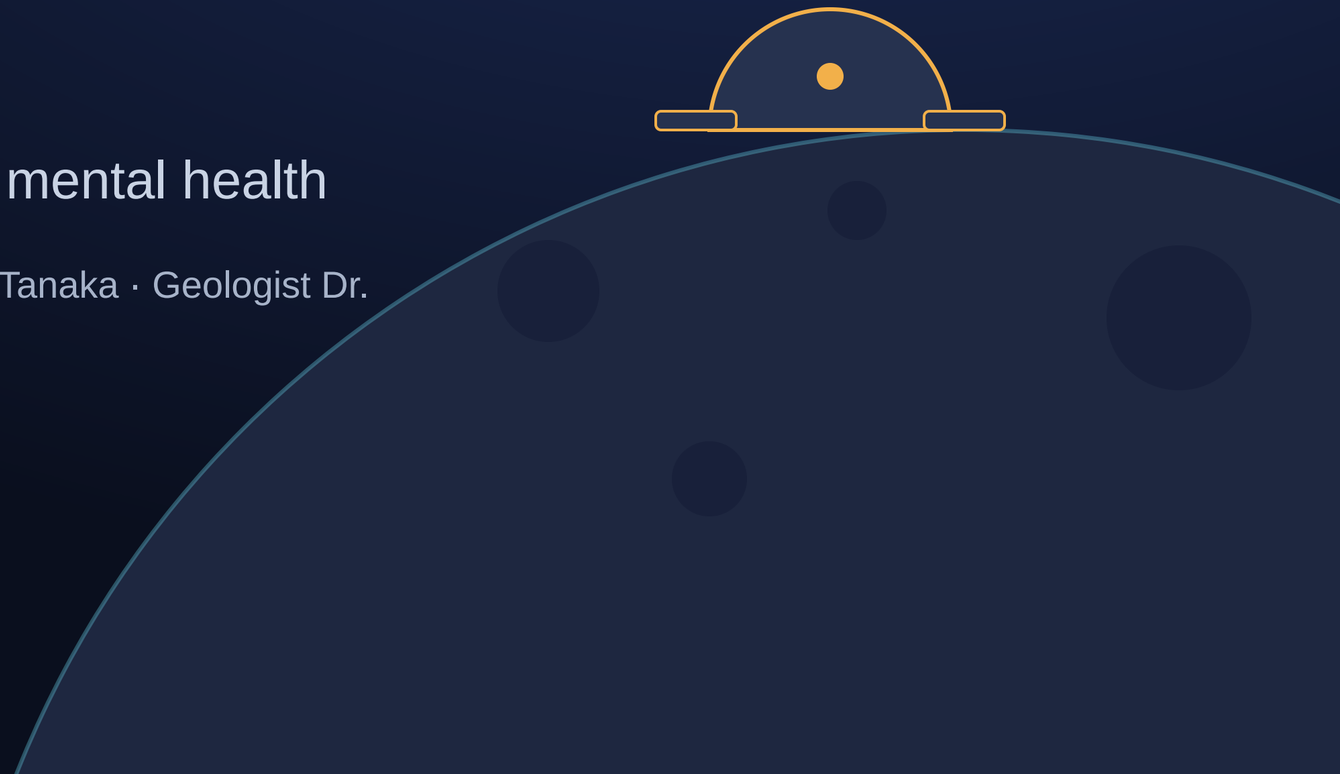
NASA HUNCH · PROJECT ARES · AGENCY REVIEW BOARD

Lunar Smart Adaptive Habitat

AI psychological monitoring for positive crew mental health

Crew of four · Cmdr. J. Rana · CMO Dr. L. Okafor · Engineer K. Tanaka · Geologist Dr. S. Rivera



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THE PROBLEM

Six conditions threaten an isolated, confined crew

Anxiety

Acute spikes. Must persist 3 min · monitored 20 min

Lever: calm

Depression

Slow-building. Must persist 48 h · monitored 72 h

Lever: activate

Interpersonal conflict

Between a pair. Must persist 6 h · monitored 24 h

Lever: separate

Cognitive fatigue

Long focused work. Must persist 2 h · monitored 4 h

Lever: alert

Autonomy deprivation

Over-control. Must persist 24 h · monitored 48 h

Lever: give choice

Sensory monotony

Habituation. Must persist 48 h · monitored 72 h

Lever: novelty

Long-duration analogues (Mars-500, HI-SEAS, Antarctic winter-overs) report friction peaking in the second half of missions — the “third-quarter” effect.

DESIGN PRINCIPLES

Three rules shaped every decision

01

The least sensing that still keeps the crew safe

No cameras or mics in private pods. Voice and face are processed on the device; only 4 numbers a minute leave it.

02

Autonomy is non-negotiable

“No” stops a support at once and the AI waits 6 h. Only a human may follow up — the AI never overrides a refusal.

03

Change the environment, not send messages

Light, sound, temperature, scent and schedule — reversible in seconds, fading back over 30 min. Helps without exposing anyone.

D1 · SENSOR-TO-CONDITION MATRIX

Nine sensors, each earning its mass

Sensor	Placement	Qty	kg each	W each	Feeds
Heart-rate variability	Beacon + wristband	4	0.03	0.05	4 cond.
Galvanic skin response	Beacon (chest)	4	0.02	0.05	1 cond.
Sleep architecture	Wristband (night)	4	0.03	0.05	3 cond.
Voice prosody	Shared-zone mic arrays	8	0.25	2.5	6 cond.
Facial action coding	Shared-zone cameras	7	0.35	4.0	5 cond.
Movement / proximity	UWB beacons + doors	28	0.15	1.2	6 cond.
Cortisol patch	Skin patch, 24–48 h	4	0.01	0.01	1 cond.
CO2 / air quality	Every zone, in pairs	34	0.12	0.8	1 cond.
Task performance	Lab/command terminals	0	0	0	3 cond.

Installed mass

13.1 kg

≈ $13 M at ~$1 M/kg

Installed power

110 W

≈ 0.3% of the 40 kW fission supply

+4.8 kg cortisol patches for 180 days. About half the hardware is shared with comms, safety cameras and ECLSS.

Source: SensorSpec in the simulator · thresholds vs each person’s own 14-day baseline

D1 · FALSE POSITIVES & GRACEFUL DEGRADATION

A raised heart rate on the treadmill is not anxiety

Avoiding false alarms

- **Context first:** HRV, GSR and voice weigh ×0.25 during exercise or EVA and 30 min after
- **No single-sensor decisions:** each score blends 4–6 sensors
- **Personal baselines:** deviation from your own 14 days
- **Smoothing + persistence** before any action
- **Steep tiers:** T1 0.35 · T2 0.55 · T3 0.75

When a sensor fails

- **Degraded:** its weight is halved, extra noise assumed
- **Failed:** removed; weights renormalise; confidence drops
- **Confidence under 50%:** capped at Tier 1
- Every condition has at least 3 independent sensors
- Life-safety alarms are a separate, human-owned system

Rule: silently losing a detection capability is safer than a false alarm.

D2 · HABITAT SPATIAL DESIGN

Six ARES zones, each with a psychological job

Communal

Meals, meetings, game nights; first target for reconnection. 2700–3000 K, 22 °C.

Private retreat

The one place nobody watches. Door sensor only — no camera, no mic. 1800–6500 K, you choose.

Work & mission bay

Long focused work. 6000–6500 K, 21 °C, CO2 below 600 ppm, airflow 0.3 m/s.

Exercise bay

Treats low mood, tension, fatigue; protects bone in 1/6 g. 5500–6000 K, 19 °C.

Nature & growth

Tending and harvesting as behavioural activation. Red/blue grow LEDs, 55% RH, birdsong.

Transition corridors

Light and temperature blend between zones — a cue that you have “arrived somewhere”.

7 face-coding cameras (shared only) · **8** mic arrays · **34** CO2 sensors (pairs) · **28** motion/door · **17** tunable lighting panels · **12** speakers · **10** scent dispensers · **6** airflow vents

Sensors + control nodes counted in the simulator habitat · 4 crew wearables

D2 · TWO CONFIGURATIONS

Config B creates space without isolating anyone

Element	Config A — nominal	Config B — conflict protocol
Meals	One shared wardroom table	2nd dining spot: fold-out table in the Lounge
Wardroom	Open plan	Acoustic screens N + S on a ceiling track
Lab	Shared workbench	Privacy screen between workstations
Routes	One main route through the core	Lit alternate route on the lower deck
Commons	Games / movies	Warm “café” second social space
Mediation	—	Opt-in private check-ins

Why it works

Natural separation and de-escalation paths, no one told who is in conflict. Reversible in minutes, no tools.

Who deploys it

Never the AI alone: a crew member accepts the Tier-2 offer, or the Commander switches it (key C).

D3 · SEVERITY TRIAGE

One score per condition, three tiers



Weighted, smoothed sensor score 0–1 · up to +0.05 for a person whose normal is higher

Tier 1 · mild

Silent, subtle, reversible environment changes. Not announced — but always in the person's own log.

Silent ≠ secret

Tier 2 · moderate

Stronger changes + a gentle offer in plain words. "No" stops it; the AI waits 6 h.

No answer in 1 h = "not now"

Tier 3 · severe

The person is told first, then the CMO. Tier-2 support continues while humans take over.

Backup: Commander → Mission Control

Can only LOWER the tier

Not persisted long enough → Tier 0 · confidence under 50% → max T1 · weekly review overdue → max T1 · crew paused or said no → AI holds back

Can RAISE it

Support not working by the end of its window → up one tier

D3 · FLOWCHART 1 OF 6 · NINE STAGES

Anxiety: calm within a 20-minute window

① Detection inputs
HRV 30 · GSR 25 · voice 20 · sleep 10 · face 10 · movement 5 (%)

② Signal weighting
No sensor above 30%. HRV, GSR, voice ×0.25 during and 30 min after exercise/EVA

③ Duration
Smoothing τ 0.1 h; must stay above threshold for 3 min

④ Severity triage
≥ 0.35 T1 · ≥ 0.55 T2 · ≥ 0.75 T3, then the lowering rules

⑤ Interventions
T1: 2400 K amber, alpha 10 Hz, 23 °C, lavender. T2: VR retreat offer

⑥ Notification & consent
T1 silent but logged. T2 Yes / No. STOP, pause 24 h, AI off any time

⑦ Monitoring · 20 min
Score −15% or below T1; RMSSD within 10% of baseline; GSR under 3 peaks/min

⑧ Resolution
10-min scent purge, 30-min fade back; baseline updated 10%

⑨ Escalation
CMO must acknowledge within 15 min → else Commander → Mission Control

Evidence: slow breathing near 6 breaths/min raises HRV; warm dim light and lavender calm modestly.

D3 · FLOWCHART 2 OF 6 · NINE STAGES

Depression: activate, don’t just calm

① Detection inputs
Sleep 25 · voice 20 · movement 20 · face 15 · cortisol 15 · HRV 5 (%)

② Signal weighting
No sensor above 25%. Voluntary 2-question self-check adds a signal for 24 h

③ Duration
τ 24 h; persist 48 h. Fast lane: poor sleep AND low activity → T2 after 36 h

④ Severity triage
≥ 0.35 T1 · ≥ 0.55 T2 · ≥ 0.75 T3, then the lowering rules

⑤ Interventions
T1: morning bright light, greenhouse spectrum. T2: tend/harvest, exercise, family call

⑥ Notification & consent
T1 silent but logged. T2 is an invitation: Yes / No / ignore

⑦ Monitoring · 72 h
Sleep efficiency ≥ 80%, active + social minutes rising, voice energy toward baseline

⑧ Resolution
Below T1 → fade back over 30 min; episode logged; baseline updated

⑨ Escalation
CMO must acknowledge within 1 h → else Commander → Mission Control

Evidence: behavioural activation and exercise treat depression; morning bright light (≈10 000 lux) helps mood.

D3 · ENVIRONMENTAL LEVERS

Six levers, used least-noticeable first

Lighting
1800–6500 K, 0–10 000 lux · warm ≈2200 K after 20:00 · dawn simulation

Audio
Delta 2 Hz sleep · alpha 10 Hz calm · gamma 40 Hz focus · nature beds

Thermal
18–26 °C · 18–19 °C sleep · 23 °C anxiety · 1–2 °C zone gradients

Scent
Lavender calm · peppermint alert · rotate 15–45 min · 10-min purge

Air
CO2 under 600 ppm in work bays · 45–55% RH · 0.1–0.4 m/s airflow

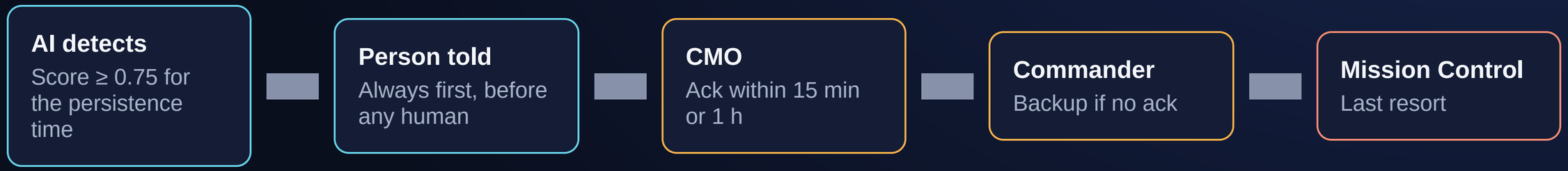
Social & routine
Family-call slots · shared meals · team task rebalancing

1 · Quietest lever where you already are → **2 · A zone you can choose to visit** → **3 · An offer in plain words** → **4 · People (the CMO)**

Two conditions at once get different zones, so levers never fight. A safety alarm overrides every lever instantly.

D3 · ESCALATION PATHWAY

The person hears it first, then a human takes over



Tier	Maximum response time	Applies to
Tier 1	Under 1 min (automatic)	All six conditions
Tier 2 offer	Under 5 min	All six conditions
Tier 3 · acute	Acknowledged in 15 min	Anxiety, cognitive fatigue
Tier 3 · slow-building	Acknowledged in 1 h	Depression, conflict, autonomy, monotony

Mission Control is told directly only if no crew responder exists or Tier 3 lasts over 24 h — condition, tier and times only, never raw data.

D4 · ETHICS AND LIMITS

Four tensions, each with a design answer

1 · Surveillance vs. privacy

No camera or mic in pods, quarters, hygiene, VR retreat or call booth. Raw audio/video never stored; logs auto-delete after 30 days; you can export or delete yours.

2 · Autonomy vs. intervention

A refusal stops the AI for at least 6 h; only a human may follow up. Agency test: no silent changes for someone already over-controlled; max 3 per 24 h.

3 · Transparency vs. distraction

Tier 1 is silent but always in your own log, with reasons and confidence. “Tell me every change” on request. Offers come once and expire after 1 h.

4 · Individual vs. crew

Act on environment and schedule, not information. The CMO decides the minimum the Commander needs: “off EVA rotation this week”, not why.

Ethics shaped the engineering: keeping sensors out of private rooms also saved ≈ 5 kg and ≈ 50 W.

D4 · HARD LIMITS & HUMAN OVERRIDE

Eight limits enforced in code — and an off switch

1. Never share one crew member’s psych data with another without consent.
2. Never override a refusal — only a human (CMO) may follow up.
3. Never touch medication, medical protocols or life-safety systems.
4. Never alert Mission Control about someone without telling them first.
5. Only reversible, crew-overridable changes (light, sound, ±2 °C, scent, offers).
6. No cameras or mics in pods, hygiene, lavatories, VR retreat, call booth.
7. At most 7 days without a human system review — else Tier 1 only.
8. Public indicators never reveal mood unless the wearer chooses to share.

HUMAN OVERRIDE

AI OFF switch

Any crew member, any time, one tap (key O). The AI may change nothing; systems return to normal.

For yourself: STOP · pause 24 h · pause monitoring 72 h · restore my room · “that reading is wrong” · delete my history.

Automatic triggers: Tier 3 → CMO takes over · no ack → next human · review missed or confidence under 50% → Tier 1 only · any safety alarm → every comfort lever stops.

LIVE DEMO · UNITY VR + DESKTOP SIMULATOR

The design is the code: nine keys, in order

F11	Deliverables D1–D4 with live numbers
C	Switch Config A ↔ B
Tab	Crew interface: AI-actions log, privacy
F2	CMO console: acknowledge a Tier 3
O	AI off switch — everything returns to normal
M	Annotated floor plan with coverage rings
F1	Dashboard: inject a sensor failure
F5	Board scenarios Q1–Q6, run live
F4	72-hour tabletop on the real engine

Tabletop script: Rivera’s slow depression, an Okafor–Tanaka conflict, a cortisol-patch failure and a 4-hour CMO EVA — with daily exercise as the false-alarm control.

AGENCY REVIEW BOARD · Q&A PREP

Six board questions, six short answers

Q1 · Workout false positive

HRV, GSR, voice ×0.25 for 30 min after exercise/EVA keeps the score below T1; persistence catches real anxiety.

Q2 · Depression + conflict

Parallel protocols, own clocks (72 h / 24 h). Greenhouse + wardroom vs. commons + timing; Config B on acceptance.

Q3 · Silent change found

Pre-mission consent + personal log. Recovery: apologise, show the log, offer “tell me every change”, CMO if wanted.

Q4 · Cortisol fails Day 45

Weights renormalise; under 50% confidence → T1 max. Daily self-check, weekly CMO chat, saliva test card to Day 90.

Q5 · No permanent opt-out

Every psych sensor opts out; 72-h pause. HRV floor + CO2 stay on (vital signs). Permanent: via CMO talk.

Q6 · CMO on 4-h EVA

Commander backs up + suit ping. No ack in 15 min / 1 h → Mission Control, after the person is told. T2 continues.

Each answer runs live in the simulator: press F5 and pick the question.

THANK YOU

Questions from the board?



Every decision — technical, spatial, procedural and ethical — aims to keep four people psychologically healthy on the Moon.

Jaineil R. — design, modelling, software

Angel C. — research & requirements

Farhan K. — physical model

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